

```
$losetup -a | tail -1
```

After creating the loop device, it's time to format it. You can use any file system, but I recommend using the `ext2` file system (the reason will be explained later).

To format, use `mkfs utils`:

```
$mkfs.ext2 /dev/loop0
```

If you are working on big data management, then the `xfs` file system is preferred, because it works well on handling large files and supports larger inode data. Now you can mount it to use as your virtual storage, which can be shared, as follows:

```
$mount /dev/loop0 /mnt/
```

To share it across the network using a service like `nfs`, open `/etc/exports` and use the following command:

```
/mnt/ *(rw, rsync, no_root_squash)
```

Close it and restart the `nfs` service by using the command given below:

```
$service nfsd restart
```

Once you put the data inside the device, you can check its original occupied size by using the `qemu-img` utility, as follows:

```
$qemu-img info test.img
```

...or by using `du`, as shown below:

```
$du -h test.img
```

General disk operation

Scaling up the size: If a user wants more storage, it can get difficult for you since this type of storage doesn't support disk operations like resizing (especially online resizing) and you can't unmount or put the storage offline. Using a small trick and with a bit of manipulation, however, you can achieve this. Just follow the exact procedure I have outlined. Suppose a user wants his space to be increased by 1GB (i.e., 2GB total), then you first need to increase the size of the file by 1GB. For that you can use the same `truncate` tool, since it works on already created files and also creates the file of the desired size.

```
$truncate --size=2GB test.img
```

...or:

```
$qemu-img resize test.img +2GB
```

You will find that there is no change in the mounted space, since the loop device has not detected the increased size. To make the loop device detect this, it will first have to be checked by `e2fsck`, using the following command:

```
$e2fsck -f test.img
```

After that, since the file system is only made on a 1GB partition, it has to be resized. However, we don't want data to be lost or deleted and can't handle the break in service by unmounting it for even a few seconds. So we need to do online resizing by using the command given below:

```
$resize2fs test.img
```

We have specially used the `ext2` file system, since online resizing and file checking only works perfectly for this file system with `resize2fs` utility. This will make our file partition ready for use. However, all these changes have not yet been detected by the loop device; so the same operations have to be done on it.

```
$losetup -a |grep mnt
```

```
$losetup -c /dev/loop0
```

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